3 Counting, Plurality, and Portions

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1. Introduction

The goal of this paper is to explore the relation between plurality, discreteness and countability. It is often taken for granted that the singular–plural distinction is relevant only for count nouns, since mass nouns do not have natural plural readings where they denote pluralities of individuals, as illustrated for English in (1):

(1) a. Cats are lying on the floor.
 b. #Muds are lying on the floor.

(1a) is felicitous with a natural existential reading where the plural gives information about quantity. The sentence is true if at least one and possibly more individuals which are cats are lying on the floor. (1b) does not have this

[Editorial note (all editorial notes by Fred Landman)] This paper is an edited version of the last version that Susan wrote, a manuscript finished a month before she was first taken into the hospital, and three months before she died. The file was called "provisional final version". She had told me that she thought very little needed to be done still, but since we were both extremely busy at the time, she didn't go into details (except for some points noted in footnotes below). Normally, she would have made those changes, given it to me for editing and a final round of comments, and then sent it off. As it is, this time I had to do the editing one stage earlier and not long after she died. I have edited all Susan's papers and books in this way since 1992 (she hated proof reading, and was happy with me suggesting little bits of formal cleaning up that she didn't have the patience for), and I can say that the paper I worked on was actually finished. I didn't have to do more than I would usually do anyway; in fact, less, because in other papers I might even at this stage have substantial comments to make which would have led to changes, but I did not have such comments for this paper, which I think is a lovely, mature Rothstein paper. As usual, editing Susan's papers is hearing her voice coming up from the page and entering into a dialogue, which at this stage was moving in its own way.

[Editorial note] Susan did not write acknowledgements for this paper. She would have thanked the audiences where she presented this material, the students and colleagues in the weekly meetings of her research group at Bar-Ilan; she would have thanked me in a witty way; but her greatest thanks would have been for the researchers that took the questionnaire that she and Suzi Lima designed to the communities in the Amazon region where they did their fieldwork, who joined them at the conference where the results were discussed, and who wrote papers for Lima and Rothstein (2020). And among them, she would have had no higher praise than for Suzi Lima. This paper owes a lot to Suzi's work, their collaboration, and the many conversions they had, and to Suzi herself, who very soon after our first meeting became a dear friend of both of us.

interpretation. To the degree that it is felicitous, it can only be interpreted as meaning that instances of different kinds of mud are lying on the floor: the plurality gives information about the multiplicity of kinds and not instantiations.[1] This reading is not a quantity reading, since "I collected few muds/a lot of muds" tells you nothing about the quantity of mud collected, while "few muds" can be exemplified by a lot of mud, and "many/a lot of muds" can be exemplified by what amounts to very little mud. These readings raise a set of independent issues, and since I will focus on quantity interpretations of noun denotations and the contribution of plurality to these quantity interpretations, I will ignore kind readings in this paper.

The contrast in (1) correlates with the fact that count nouns can be modified by numerals while mass nouns cannot. In English, count nouns modified by *one* are in the singular form while count nouns modified by *two* and above are in the plural. In (2a), the numeral gives information about the quantity of cats on the sofa. (2b) is infelicitous without heavy contextual support indicating why it is plausible to talk about types of mud, and the numeral does not give any information about the quantity of mud on the sofa, only about the number of types of mud. A mass noun can be combined with a numeral only when a classifier or expression indicating a measure unit is used. In the counting expressions in (2c), *puddle* and *item* characterise the individual portions of mud and furniture which are being counted.[2]

(2)	a.	There is one/a cat on the sofa.	There are two/three/many cats on the sofa.
	b.	#There is one/a mud on the sofa.	#There are two/three/many muds on the sofa.
	c.	There is one puddle of mud.	There are two items of furniture in the room.

There are cases like (3), where a mass noun does pluralize and can be modified by a numeral.

(3) We ordered here two coffees, three beers and one water.

[1] [Editorial note] This is a paragraph that Susan wanted to reformulate in the light of our discussions concerning attested examples like (i), from my then-forthcoming book (Landman 2020), where the available reading of "mud" does not involve different *kinds* of mud, but different *samples* of mud:

(i) Edzwald and O'Melia's analyses of *three muds* from the Pamlico estuary (N.C.) indicate the stability of *the natural muds*
(. . .) is lowest in the upstream, fresh water part of the estuary.

[2] Rothstein (2009, 2011, 2016, 2017) argues that individual classifiers such as those in (2c) and mensural classifiers such as *kilo*, *liter* etc. are of different types and interact with numerals in different ways.

These are considered to be standardized portion readings. The noun seems to have shifted from a mass noun to a count noun because of the specific "restaurant context".

It is often assumed (e.g. Chierchia 1998a, 2010) that plurality and countability are necessarily related. One way of expressing this might be as follows. Counting the number of cats involves determining how many individual cats there are. "There are four cats on the sofa" is true if the plural individual on the sofa is made up of four discrete, non-overlapping single individual cats. It is thus natural that the set of individual cats is grammatically salient and marked in contrast to the plural set. Since we do not count mud, and mud does not come in inherently individuable stable units, there is no set denoting single units of mud and the singular–plural contrast is not relevant. In (3), the plural marker indicates that the noun has shifted to a count interpretation. Theories differ as to why *mud* is mass and why, unlike *coffee* or *water*, it cannot shift to a count interpretation, but the assumption that the lack of countability and the lack of pluralization are related is pretty widely assumed, as we will see below.

The goal of this paper is to explore the relation between countability and plurality, especially in examples like (3). There are two possible analyses of these examples. One is that the noun has shifted to a count noun, and therefore pluralizes, and the second is that pluralization has in some sense coerced the mass noun into a count noun reading. As we will see, these two analyses are not equivalent, since the second requires a plurality operation to presuppose a set of discrete individuals on which it can operate. We shall show that plurality in English is not presuppositional in this way, but suggest that in other languages it might well be. We shall further show that plurality and countability are not necessarily dependent on each other: countable nouns are not necessarily marked plural and plural nouns are not necessarily countable.

The structure of the paper is as follows. We begin with reviewing basic properties of the count–mass distinction (Section 2) and of the semantics of plurality (Section 3). The next section of the paper reviews cases of pluralization of mass nouns in two count–mass languages, and proposes a semantic analysis of the data. We then turn to data from a recent fieldwork project on indigenous languages of Brazil which suggests that the semantic properties of pluralization may vary from language to language. These data together suggest that an account of the relation between plurality, discreteness and countability must be much more nuanced and allow for more variation than has previously been assumed.

2. The Count–Mass Distinction

As is well known, the count–mass distinction is a distinction between nouns which can and nouns which cannot be directly modified by numerals, as in (4), and "count–mass languages" is a term used to describe languages which have a count–mass distinction among numerals, for example, English.

(4) a. three cats b. #three sands; #three muds; #three furnitures

A count–mass distinction is usually associated with other grammatical properties. For example, in English, count nouns usually display a singular–plural contrast, independent of the presence of numerals; as indicated in (5), count nouns are modified by *many* and *few*, while mass nouns are modified by *much* and *little*, and count nouns can be the complement of *every*, while mass nouns cannot:

(5) a. many cats, #much cat(s) b. #many muds, much mud
#many furnitures, much furniture
c. every cat d. #every mud; #every furniture

Cross-linguistic variation with respect to the count–mass contrast occurs along at least three different parameters. First, languages with a count–mass distinction may differ as to which nouns are mass and which are count. For example, *furniture* is a mass noun in English, while in Modern Hebrew, the corresponding nominal has both a mass form *rihut*, which translates as "furniture", and a count form *rehit*$_{sg}$/*rehit-im*$_{pl}$, which best translates as 'item/items of furniture'. Second, count–mass languages may differ as to the grammatical properties which characterize mass and count nouns. For example, while in English one asks "How many Ns?" when N is count and "How much N?" when N is mass, Modern Hebrew uses *kama* in both cases. Thirdly, not all languages have a count–mass distinction. For example, in Mandarin no noun can be directly modified by a numeral. All nouns, including those which are typically "count", require a classifier when combining with a numeral, parallel to the English mass nouns in (2c):

(6) a. sān *(zhī) gǒu
three Cl$_{small\ animal}$ dog
"three dogs"

b. sān *(píng) jiǔ
three Cl$_{bottle}$ wine
"three bottles of wine"

At the other extreme, Lima (2014a) shows that there are languages, *in casu* Yudja, in which all nouns can be directly modified by a numeral, as shown in (7). In (7a) individual pacas (a small rodent) are counted, while in (7b) portions of flour, in context most plausibly bags of flour, are counted:

(7) a. Txabïu ba'ï wãnã
three paca ran
"three pacas ran"

b. Txabïu asa he wï he
three flour in port in
"there are three (bags of/quantities of) flour in the port."

3. Plurality

The by-now classical theory of plurality, which I will adopt here, is based on Link (1983). Singular count nouns denote sets of individuals, and plural count

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nouns denote Boolean (semi-)lattices, i.e. the denotations of the corresponding singular nouns closed under sum.[3] Plural count nouns are thus cumulative in the sense of Krifka (1989).

(8) Pluralization
$*P= \{x \in D: \exists Z \subseteq P: x = \sqcup Z\}$
i.e. the plural operator applied to the set X yields the closure of X under sum.

(9) $PL(\{a, b, c\}) = *\{a,b,c\} = \{a, b, c, a\sqcup b, a\sqcup c, b\sqcup c, a\sqcup b\sqcup c\}$

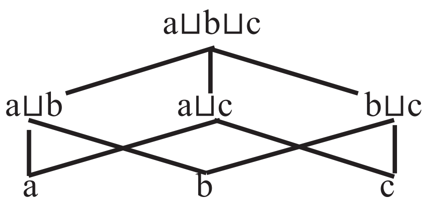

Fig. 3.1: Link-style atomic semi-lattice of plurality

In English, and in many other languages, singular individuals are part of the plural denotation, thus accounting for examples like (10a) as well as "at least one" readings as in (10b):[4]

(10) a. Do you have grandchildren? Yes, one grandson.
b. I have had students who did not matriculate in maths. At least one.

The plural morpheme marks the operation in (8).

(11) $BOY = \{a,b,c\}$; $*BOY = \{a, b, c, a\sqcup b, a\sqcup c, b\sqcup c, a\sqcup b\sqcup c\}$

In English-type languages, then, the singular–plural distinction marks the contrast between sets of entities with a cardinality of "1" and sets of entities of non-determined cardinality. In mass noun denotations, no such distinction is made. Mass noun denotations are assumed to be closed under sum, i.e. to be inherently plural, or inherently cumulative, and thus plural marking, indicating closure under sum, is irrelevant and/or redundant (Chierchia 1998a). The

[3] Link (1983) treats plural count nouns as denoting semi-lattices. Landman (2010) argues convincingly that they denote complete lattices. For our purposes here, this distinction does not matter. While the operation given is the complete lattice operation that adds the minimum 0 to the closure under sum, I will ignore 0 in this paper.

[4] There is no reason why all languages should behave like this, and in fact Bale and Khanjian (2014) argue that in Western Armenian, the unmarked form of the noun denotes the set of individuals closed under sum, while the plural marked noun denotes the set of plural individuals minus the atomic elements.

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association between count nouns and plurality is thus assumed to be fundamental. As Chierchia (2010: 108) writes:

the mass/count contrast in number marking languages is typically centred on the distribution of singular versus plural morphology … in number marking languages the distinction affects the distributions of plural versus singular morphemes.

There is, however, a lot of cross-linguistic variation in the expression of plurality. For example, in Dëne Sųłiné, a Northern Athabaskan language spoken in northwest Canada, there is a count–mass distinction, but no marking of plurality (Wilhelm 2008).

(12) a. *sôlághe dzóå* b. **sôlághe thay*
Five ball five sand
'five balls'

Thus, the grammatical expression of a count–mass distinction does not necessarily require a morphological contrast between singular and plural count nouns.

In the next section, we will examine examples of the converse situation, i.e. languages which have a count–mass distinction where the singular–plural distinction shows up on many mass nouns as well as on count nouns. The examples will be taken mainly from English, with some reference to Modern Hebrew.

4. Pluralization of Mass Nouns in a Count–Mass Language

Cases of pluralization of mass nouns can be divided into two kinds: (i) cases where the mass noun has shifted to a count noun, and has acquired the ability to pluralize along with all the other properties of count nouns, and (ii) cases where the mass noun can pluralize but has not shifted to a count interpretation. We will look at each in turn.[5]

4.1. Plural Mass Nouns which Shift to Count

English has a range of nouns which are flexible and can appear in both mass and count constructions, such as *stone* and *brick* in *not much stone* vs. *many stones*, *a lot of brick* vs. *three bricks*. The shift from mass to count is apparently free, in the sense that there is no contextual restriction. The count

[5] Two other cases which we will not discuss here are: (i) *pluralia tantum* such as *scissors*, *trousers*, *alms* (English) and *mayim* "water", *šamayim* "heavens", *ofanayim* "bicycle" in Hebrew, though note that these have formally dual endings rather than plural; (ii) plurals of kinds such as *my three favourite Scotch whiskies*. As discussed above, these are not quantity readings.

noun denotes a concrete portion of the substance denoted by the mass noun. In some cases, such as *stone*, there are no restrictions on what the portions are. In other cases, what counts as a portion in the denotation of the count noun is highly restricted. While stones may be just pieces of stone, bricks are chunks of brick of a standardized shape and size. There is a well-defined subgroup of these nouns where the mass noun denotes a substance which is eaten or drunk, and the count noun denotes a standardized portion of this substance, as in *a lot of coffee* vs. *three coffees*. Bare plural readings and definite expressions are also available with an existential interpretation.

(13) a. The waiter handed out beers and coca colas to all the guests.
b. He handed out the coffees and the beers to the people who had ordered them.

While these nouns are flexible, they can only be used in restaurant contexts, and we refer to them as *horeca* nouns (following the Dutch mass noun *horeca* standing for ***ho***tels–***re***staurants–***ca***fés).

Note that flexible nouns, such as those in (13), have all the other properties associated with count nouns, in particular they occur with numerals, with *each* and *every*, and with reciprocals.

(14) a. Three coffees, two waters; three porridges
b. Every beer you drink will cost you five Euros.
c. Each coffee is brewed separately.
d. I see that our coffees are standing next to each other on the bar getting cold.

There are two possible explanations for examples like (13). One possibility is that the mass noun N shifts to a singular count noun interpretation, where it denotes a set of discrete standardized portions of N, and then, as a count noun, pluralization applies in the normal way. The other possibility is to analyse (13) as the result of coercion.

Pluralization is an operation which applies to singular count noun denotations and gives the closure of a set of discrete entities under sum. It can thus be taken to presuppose a set of contextually discrete atoms as an input. When applied to a mass noun, the presuppositions of the plural operator are accommodated, and the mass noun is coerced into an interpretation in which it denotes a set of contextually discrete entities, in this case standardized portions, closed under sum. A similar explanation of coercion would account for (14).

However, the existence of a large range of examples where mass nouns pluralize but the resulting nouns do not denote sets of discrete entities closed under sum and do not have all the properties of count nouns suggests that plurality is not presuppositional in the way just outlined. We look at these examples in the next sub-section.

4.2. *Plural Mass Nouns which Clearly Haven't Shifted to Count*

There are at least three kinds of mass nouns which pluralize but show (almost) none of the other properties of count nouns.

First, plurals of mass nouns may denote **multiplicities of events**. As Moltmann (1997) notes, plural nouns such as *rains* can be modified by adjectives expressing multiplicities, as in *frequent rains*, although in other respects *rain* behaves like a mass noun, as is shown by the infelicity of (15):

(15) How many rains were there this winter?[6]

While weather mass nouns naturally occur in the plural with eventive readings, the phenomenon occurs with other nouns too, as in (16):

(16) a. He has *depressions* very often.
b. #He has had *three depressions* this year.
c. He has had *three events/periods of depression* this year.

Epstein-Naveh (2015) explores the phenomenon in Hebrew, showing that *gešem* "rain" can occur in the plural and be modified by a frequency adjective. The example in (17) shows clearly that the plural *gšamim* "rains" denotes events of rain and is not to be interpreted as an abundance plural of the type we will discuss shortly.

(17)

me'at	miška-im	še-megi'im	be-cura	bilti	sdira
little	precipitation-PL	that-come-PL	in-form	not	regular
šel	*gšam-im*	*nedir-im*	ve-šitfon-ot . . .		
of	rains.PL	rare-PL	and-flood-PL . . .		

"the little precipitation that come irregularly in the form of infrequent rains and floods"

As (17) shows, these plural nouns can be modified by frequency adjectives which must agree in plurality with the noun. However, unlike the examples in (13) and (14), these nouns show no other properties of count nouns. For instance, numeral modification is impossible.

(18) a.

#šlosa	gešam-im	yardu	etmol	/ha-xoref
three	rain.PL	fell.PL	yesterday/this	winter

Intended: "there were three events of rain yesterday/this winter"

b.

#gešem	exad	yarad	etmol
rain.SG	one	fell.SG	yesterday

Intended: "One rain fell yesterday/It rained yesterday"

[6] I found one example on the web of *rains* used as true count noun, namely in a scientific article entitled "Distinguishing in a puddle the water from two rains; a crucial methodological issue", www.ncbi.nlm.nih.gov/pubmed/21300567.

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Again, the phenomenon is not restricted to weather nouns and Epstein-Naveh identifies a number of other pluralized mass nouns that may have eventive interpretations. As well as other weather nouns such as *šeleg* "snow" vs. *šelagim* "events of snow occurring", these included *nezeq* "damage" vs. *nezaqim* "events of damage occurring" and *laxac* "pressure" vs. *lexacim* "events of pressure being exerted".[7] Ghaniabadi (2012) mentions similar cases in Persian.

Next we have **"abundance" plurals**. Plurals of mass nouns have been identified in uses which express excess or large quantities, as in (19) (Corbett 2000, using the phrase in Cowell 1964; see also Acquaviva 2008).

(19) a. "Snow looks gorgeous on your fountain – mine is just full of water and fallen leaves as *the rains are heavy today and the winds gusting to 30 mph.*" (http://willows95988.typepad.com/tongue_cheek/2009/01/snowing-in-provence-.html)
b. "Elgar's life was the time when crowds would wait on the white cliffs of Dover to cheer home a line of ships, with names like HMS Formidable, their hulls pulverising the waters, as they advanced *through the fogs of the English Channel*" (www.bmj.com/content/341/bmj.c6965/rr/686305)

Note that the temporal adverb *today* in (19a) and the episodic verb in (19b) mean that *rains* and *fogs* cannot be taken as denoting a plurality of events. Instead, these nouns seem to denote pluralities of concrete instantiations or portions, but, crucially, these concrete portions are not discrete and are certainly not countable.[8] Abundance plurals are widespread: Alexiadou (2011) and Tsoulas (2009) report on abundance plurals in Greek, Ghaniabadi (2012) documents them in Persian, and Doron and Müller (2014) in Modern Hebrew.[9]

[7] Acquaviva (2008: 110) quotes an example from Lucretius (*De Rerum Natura* 5.25) suggesting a similar usage with the plural of the Latin count noun *solibus* 'suns', given below in (i). He suggests that the plural noun refers to "sun stages as ever-returning sun-events". Note also a similar usage in Rupert Brooke's poem *The Soldier* in (ii):

(i) Pars terrain nonnulla, perusta **solibus** adsiduis
"a large part of the earth, scorched by continuous suns"
(ii) "Washed by the rivers, blest by suns of home."

[8] Note that *miška-im* in example (16), which appears with the quantity expression "a little" is an example of *pluralia tantum*.

[9] [Editorial note] One of the most beautiful examples of a plural of abundance is *veel waters – a lot of waters*, as it occurs in the translation of Numbers 20.11 in the Dutch Statenvertaling of 1637, when compared with the same paragraph in the English Geneva Bible of 1560:

(i) Toen hief Mozes zijn hand op, en hij sloeg de steenrots tweemaal met zijn staf; *en er kwam veel waters uit*, zodat de vergadering dronk, en haar beesten.
(ii) Then Moses lift up his hande, and with his rod he smote the rock twise, *and the water came out aboundantly*; so the Congregation and their beastes dranke.

I include this example here, because Susan would have loved to do so.

Thirdly, we have **plurals of abstract mass nouns whose denotation denotes multiplicity of instances** (Acquaviva 2008; Grimm 2016). Abstract nouns are often pluralized to denote a multiplicity of instances, but like eventive and abundance plurals, these do not have the other properties of count nouns, as shown in (20). Although some cases of these nouns may be found with numerals, speakers intuitions are that they are infelicitous or highly marked:

(20) a. "The *affections* are more reticent than the passions, and their expression more subtle." (EM Forster: Howard's End)
b. #My *three affections* are. . .

Similarly, *harms* occurs frequently in the plural, but *two harms* is marked.

Our generalizations, at this stage, are as follows. In general, mass nouns can pluralize without acquiring all the properties of count nouns, most specifically without being directly modifiable by numerals. In these plural uses, they do not denote sets of sums of necessarily discrete individuals or entities, although they may make reference to pluralities of possibly overlapping quantities or instantiations. There is a specific subset of mass nouns which pluralize and which do occur with numerals. These are the highly restricted portion readings which occur in *horeca* contexts, such as *three beers, two coffees*. Here it seems that the pluralized mass nouns have acquired all the properties of count nouns. Notably, in English and in Hebrew, three properties co-occur: if a pluralized mass noun can freely occur with numerals, then it also has a singular indefinite reading and its denotation involves discreteness. The singular noun occurring in a count context denotes a set of discrete entities, while the plural denotes this set closed under sum.

Let us assume, then, that when a pluralized mass noun has acquired all count properties, it has in fact shifted to count. Pluralization then indicates the application of the standard plural operator in (8) above. The question is, what is the semantic effect of pluralization in the cases where shift-to-count has *not* taken place? We will turn to this in the next section.

5. A Semantics for Plurals and Count–Mass

We assume the account of the count–mass distinction in which count nouns are derived from mass nouns (Krifka 1989; Chierchia 1998a; Rothstein 2010). In Rothstein (2010), count noun denotations are relativized to contexts. The domain of type e is the mass domain, a join semi-lattice, and root nouns are interpreted at type <e,t> as sets closed under sum. The interpretation of mass nouns is identified with that of root nouns. Count nouns are derived from root nouns via an operation COUNT_k, where context k is a subset of the domain of type e, and COUNT_k applies to a root noun at type <e,t > and results in an

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expression $N_k = COUNT_k(N_{root})$ of type $<e \times k, t>$, denoting the set of k-indexed entities which count as atoms in context k. These atoms are contextually determined discrete non-overlapping entities each of which counts as "one" instance of N in context k. (See also related discussion of overlap in Landman 2011, 2016). Examples are given in (21):

(21) $[\![stone_{mass}]\!] = MASS(STONE_{root}) = STONE_{root}$
$[\![stone_{count}]\!] = COUNT_k(STONE_{root}) = \{<e,k>: e \in STONE_{root} \cap k\}$

stone$_{mass}$ denotes a set of quantities of stone; while *stone*$_k$ denotes a set of type $<e \times k, t>$, namely:

$\{<e,k>: e \in STONE_{root} \cap k\}$, i.e. the set of indexed entities which count as one in context k.

While I shall assume this account in what follows, the details are not crucial.

The plural operator, given in (8) and repeated here, when applied to N_k gives $*N_k$, plural count nouns, the closure of N_k under sum.[10]

(22) Pluralization
$*P = \{x \in D_{e\times k}: \exists Z \subseteq P: x = \sqcup_{e\times k} Z\}$
i.e. the plural operator applied to the set X yields the closure of X under sum.

When pluralization applies to N_k, the semantic effect is straightforward. The question is: what is the semantic effect of the pluralization operation when it applies to mass nouns? We have identified two different situations in the previous section: cases like *the coffees* where the plural noun has all the properties of a count noun, and the remaining cases, such as *rains*, *affections*, where the noun does not.

Examples like *the coffees* have been analysed as the result of coercion (see e.g. Pelletier 1975; Sutton and Filip 2016b; and many others). The details of how this coercion would work are rarely spelled out explicitly, but a possible analysis would go as follows. *Coffee* is a mass noun. Pluralization applies to count nouns; in other words, it presupposes a count noun input. As a result, *coffee* shifts to a count denotation in which it denotes a set of contextually determined discrete *portions* of coffee. This is the presuppositional account outlined above, and, as mentioned there, I think it is problematic. The problem is that this analysis predicts that *all* occurrences of pluralized mass should be able to shift to a count reading: just make discrete portions available in context, and there is no reason for the shift *not* to take place. As we have seen, this is not the case.

The alternative account that I favour is to assume that in cases like *the coffees*, the $COUNT_k$ operation has applied implicitly and derives a count

[10] Where the lattice structure and operations are lifted to $<e \times k,t>$ in the obvious way.

noun denoting a set of contextually determined portions, as proposed in Khrizman et al. (2015). Thus instead of assuming that these cases involve a mass noun with a reading shifted to count, I propose that these cases involve a noun that has shifted to a true count noun. On this analysis, *coffee* in this context is a flexible count noun, just like *stone* or *brick*, but more like *brick* than *stone* since the portions in the count noun denotation are standardized.

In contrast to *the coffees*, where pluralization has been applied to a count noun, I would assume that *rains* and *affections* are derived by pluralization applying to a mass noun. This non-presuppositional account of pluralization allows us to explain the contrast between *(three) coffees*, where the N is a count noun homonymous with a mass noun, and *affections* or *rains* where the pluralized noun remains mass. Of course, what is missing still is an account of the semantic effect of pluralizing a mass noun like *rain* and *affection*.

On the surface, pluralizing a mass noun should have no semantic effect at all. Since mass nouns are inherently closed under sum, pluralization applied to a mass noun is the identity function. However, the discussion in the previous section suggests that there *are* semantic effects to pluralization in the mass domain. The question then is whether pluralization is the same in the mass and the count domain, or different, and how the semantic effects are derived. We examine some possibilities.

The first option is to assume there is a single pluralization operation, as in (8) and (21), expressed by the plural morpheme. The operation applies to a singular count noun denotation and gives the closure of that set under sum. Since the plural morpheme can only apply to a morphologically singular noun, it does not apply to plural count nouns.[11]

The same operation applies to the denotations of morphologically singular mass nouns. Since the input set is closed under sum already, the output is the same as the input, but there are two effects. First, as has been argued in Carlson (1977b) and Chierchia (1998a), mass nouns are ambiguous between a kind interpretation and a reading in which they denote sets of instantiations of the kind. Thus, *mud* can denote either the kind mud or the set of instantiations of mud. Since kinds are individuals and pluralization applies to sets, the pluralized mass noun can only be interpreted as denoting a set of instantiations. The first effect of pluralization is thus disambiguation. This effect is noted in Pelletier and Schubert (1989) and Acquaviva (2008). Pelletier and Schubert suggest that mass plurals refer to “a kind viewed as a concrete sum extended in space and/or time and large enough to be a feature of the environment”, while Acquaviva observes that pluralizations such as *fogs*, *rains*, *sands* and *snows* “filter out the reading as an abstract kind”. He supports this with the following

[11] We can see this from the fact that plural nouns denoting sets of single items such as *scissors*, *trousers*, don’t have a morphologically double-plural form.

example, where a concrete instantiation of water can be referred to by either the bare singular or the mass plural, as in (23a), while the kind reading can only be expressed by the bare singular, as in (23b):

(23) a. The river discharges its water/waters into the lake.
b. The formula of water/*waters is H_2O. (Acquaviva 2008: 109)

The second effect is to focus attention on sums of instantiations. Mass nouns are non-atomic, and thus will normally contain contextually determined, but overlapping portions in their denotations. Closure under sum means that sums of these portions are also in their denotations. While such sums are not countable, they are nonetheless contextually distinguishable as sums. Pluralization draws attention to these sums, giving rise to either an abundance implication or to a plurality of instances or event instantiations. This can be thought of as following from Grice's Maxim of Manner. Since a bare singular mass noun is the simplest form of the noun, there must be a reason for using the plural form, especially in situations in which a kind reading is not plausible. Since pluralization is closure under sum, marking the mass noun as plural would draw attention to sums in the denotation of the noun N, i.e. instantiations of N which are either large (= the abundance reading) or multiplicities of instantiations (= sums of events or instances). Thus by choosing *waters* in (23a) we are drawing attention either to the large sum of water discharged into the lake or, possibly, to the fact that the water discharged into the lake is the sum of water from a number of different sources.

A second option is to try and derive the same effects semantically, rather than pragmatically. Fred Landman (p.c.) has suggested that pluralization in the mass domain might be seen as an operation which is related, but not identical, to the operation in (8) and (21). Suppose that pluralization in the mass domain is the operation which Bale, Gagnon, and Khanjian (2011) have suggested as the interpretation of the plural morpheme in Western Armenian.[12] Bale et al. argue that singular morphology in Western Armenian is associated with a number neutral interpretation of a noun N, which is an interpretation such that $N = {}^*N$, i.e. an interpretation that includes singular elements and pluralities, and that plurality forms PL(N), where PL is the operation that maps neutral interpretation N onto $N - N_{ATOM}$, which is the same as ${}^*N - N_{ATOM}$, the plural set minus the atomic elements. As we saw above, this is not the interpretation I assume for plural nouns in the count domain in English. But, plausibly, it is the effect of pluralizing a number neutral, i.e. inherently plural, set. In this case, plurals of mass nouns would denote the denotation of the N minus its minimal elements. The effect would be the same as the effect

[12] The operation itself is already in Link (1983), and there is a large literature on its semantic (dis) advantages.

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described above: it would force a non-kind interpretation of the mass noun and would force the N in context to be witnessed by *sums* of instantiations or portions, rather than by single portions, thus giving the abundance or the multiplicity of instances interpretation. On this approach, the effect would be derived semantically via the operation $*N - N_{ATOM}$, and not pragmatically.

We will not choose between these two explanations here since both converge on the same main conclusion: plurality is not presuppositional and does not entail either discreteness or countability.

We now turn our attention to the final question to be discussed in this paper: is there cross-linguistic variation in the relation between plurality, discreteness and countability, and is pluralization always non-presuppositional?

6. Cross-linguistic Variation in the Relation between Plurality and Countability

6.1. Variation

In the previous section we saw that pluralization in English is the operation of closure under sum, but does not presuppose discreteness of minimal parts. Discreteness of minimal parts is guaranteed by the count operation. Singular count nouns denote sets of contextually determined discrete minimal elements, while plural count nouns denote such sets closed under sum. Thus, while pluralization of N in the mass domain makes salient sets of sums of instantiations of N, it does not entail that the minimal parts of these sums are discrete. In English (and Hebrew) countable nouns are a subset of pluralizable nouns, and there is no presupposition of discreteness associated with pluralization.

An obvious question is whether this relation between plurality and countability is the same cross-linguistically? In this section, I shall show that the answer is no.

We have already seen that there are languages which have a count–mass distinction but no marking of plurality. The examples are repeated here for convenience (Wilhelm 2008).

(24) a. sôlághe dzóå
Five ball
"five balls"
b. *sôlághe thay
five sand
(Dëne Sųłiné)

There are also languages where all nouns behave like *coffee* in English, and mass nouns can pluralize and denote sets of pluralities of discrete entities. An example is Ojibwe (Mathieu 2012b):

(25) a. niizh baagan-a
two nut-PL
"two nuts"
b. niizh azhashki-n
two mud-PL
"two chunks of mud"

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But we will focus here on two sets of cases which are very different. One is languages where pluralizable nouns are a subset of countable nouns. The second is languages where pluralizable nouns *do* denote sets of discrete entities or portions, but these nouns are not countable. Together this suggests variability in the relation between countability and plurality.

6.2. Languages Where Pluralizable Nouns Are a Subset of Count Nouns[13]

We begin with data from Yudja. The Yudja language (family Juruna, Tupi stock) is spoken by a growing community of around 800 people,[14] most of whom live in the Xingu Indigenous Territory in the Amazonian basin in Brazil.

In Yujda all nouns can appear bare, with a singular/plural and indefinite/definite interpretation:

(26) Senahï kota ixu (Lima 2014a (03a))
man snake eat
"A/the/some man/men eat(s)/ate a/the/some snake(s)"
literally: "an undefined number of men eat(s)/ate an unspecified number of snakes"

Lima (2014a) reports that all nouns are countable, as shown in (27) and (28). Note that in neither case is there plural marking on the noun:

(27) txabïu ali
three child
"three children"

(28) a. txabïu yukïdï
three salt
"three portions of salt" (Lima 2014a)
b. Txabïu apeta pe~pe~pe
three blood drip~RED
"Three (portions of) blood dripped (in different events)"

[13] The data from Yudja are drawn from Lima (2014a, 2016), and Rothstein and Lima (2018) and were collected through a variety of methods including elicited judgements, in depth interviews and psycholinguistic experiments carried out in the framework of language documentation projects in a series of fieldwork trips by Suzi Lima between 2009 and 2016. All data not directly referenced as appearing in published works were collected by Lima on fieldtrips. Most of the speakers who are reported on come from Tuba Tuba village. The data from the other Brazilian languages cited here is drawn from research by a variety of means between 2015 and 2017 in the framework of the project "A Typology of Count, Mass and Number in Brazilian Languages", coordinated by Suzi Lima and Susan Rothstein, the results of which appear in Lima and Rothstein (2020).

[14] Siasi/Sesai, 2014 *apud* Enciplopédia Povos Indígenas no Brasil, https://pib.socioambiental.org/pt/povo/yudja.

As Lima (2014a) shows, examples like (28) are very different from English examples such as *three coffees*, which we discussed above. As discussed above, in English the counting of what appear to be mass nouns is only possible in restricted contexts, namely *horeca* contexts, and only when the portions counted are standardized portions relative to that context. If someone spills a big bottle of beer, a drop of beer and a test-tube of beer in three different places on the kitchen floor, we cannot describe this by saying "there are three beers on the floor to wipe up". In Yujda, this is not the case. Any individuable portion of a substance can be counted directly, and a big heap of salt, a grain of salt and a quantity which would fill a teaspoon together can be referred to using (28a). Similarly, if an injured person moves along a path dripping a drop of blood, a small puddle of blood and a big puddle of blood, this can be referred to as in (28b), using *txabïu apeta* "three bloods". Thus all nouns can be taken to denote pluralities of countable individuals. The individuals need not be stable through space and time, but can be individuable portions which are salient in a given context. Note that while in (28b) the portions of blood are dripped in different events, the numeral must count portions of blood and not the events. The reduplication indicating event repetition is not obligatory in (28b), and furthermore, the numeral has a special suffix when it is used to count events.

There is a plural marker, but it is restricted to [+human] nouns and it is optional. As well as occurring on bare nouns, as in (29), it can also occur on nouns modified by numerals.

(29) Senahï-**I** kota ixu
man-**PL** eat snake
"(The) men eat/ate a/the/some snake(s)"
Lit.: a plural set of men eat/ate an unspecified number of snakes

In (29), *senahï-i* must denote a true plurality (i.e. singular interpretations are ruled out). Crucially, the plural marker cannot occur on *ixu* "snake". Lima shows that this is a lexical restriction, and is not connected to the grammatical role of the noun. In relevant contexts, [+human] nouns in direct object position can be marked plural, while [−human] nouns are never marked plural, even when they are subjects.

Lima (2014a) suggests that count nouns denote maximal self-connected countable portions. Stable individuals, or natural atoms, denoted by nouns such as *senahï* "men", *ixu* "snake" are "privileged portions", and the noun does not usually denote parts of these privileged portions. But, it seems, the expression for "three bananas" *can* be used to denote three pieces of banana, three bunches of bananas, three individual bananas. The last is the preferred option, which is predicted if natural atoms have a privileged place in the ontology. Crucially, plurality does not mark countability, but seems to indicate a non-singular reading on [+human] nouns.

Yudja is not the only language in which pluralization can apply only to a subset of countable nouns. Bardagil-Mas (2020) makes the same case for Panará. The Panará data are particularly interesting since they raise the question of the presuppositions associated with pluralization.

Panará is a Jê language spoken by about 450 speakers in Mato Grosso. All nouns can occur with numerals. Pluralization, marked by the suffix *mẽra* can occur only with individual denoting nouns, while some speakers prefer to restrict plurals only to [+animate] nouns, as in (30) and (31):

(30)	rê-	s-	anpun	joopy-mẽra
	1.SG.ERG	3.SG.ABS-	saw	jaguar-PL
	"I saw jaguars"			

(31)	#rê-	s-	anpun	pakwa-mẽra
	1.SG.ERG	3.SG.ABS-	saw	banana-PL
	"I saw bananas"			

Examples like (31) can be repaired by the addition of a numeral:

(32)	rê-	s-	anpun	nõpjó	pakwa-mẽra
	1.SG.ERG	3.SG.ABS-	saw	a few/three	banana-PL
	"I saw three/a few bananas"				

Despite the fact that mass nouns may occur with numerals as in (33a), they do not usually pluralize as shown by (33b):[15]

(33)	a.	rê-	s-	anpun	**nõpjó**	mõsy
		1.SG.ERG	3.SG.ABS-	saw	a few/three	corn.
		"I saw three/a few cobs of corn/portions of corn."				
	b.	*rê-	s-	anpun	**(nõpjó)**	mõsy-**mẽra**
		1.SG.ERG	3.SG.ABS-	saw	a few/three	corn-PL
		"I saw three/a few cobs of corn/portions of corn."				

In some cases, the addition of the numeral may "repair" the sentence and license the plural:

(34)	a.	*rê-	s-	anpun	kjorinpe-mẽra	
		1.SG.ERG	3.SG.ABS-	saw	rice-PL	
		Intended: "I saw rices."				
	b.	rê-	s-	anpun	nõpjõ	kjorinpe-mẽra
		1.SG.ERG	3.SGABS-	saw	three	rice-PL
		"I saw three plates/sacks of rice."				

[15] Bardagil-Mas (2020) claims that all nouns can appear with numerals, although the data presented do not include examples of nouns denoting liquids appearing with numerals.

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Crucially, there is a possibility of pluralizing liquid-denoting nouns, but with a very specific interpretation: the plural morpheme coerces an interpretation in which the noun does not denote (portions of) the substance denoted by N, but animate individuals connected with the substance, as in (35):

(35) a. rê- s- anpun nãnpju-mẽra.
1.SG.ERG 3.SG.ABS- saw blood –PL
"I saw a few girls who were doing the menstruation ceremony."
Impossible: "I saw (portions of) bloods."

b. rê- s- anpun nõpjõ nanpju-mẽra
1.SG.ERG 3.SG.ABS- saw three blood-PL
"I saw three girls who were doing the menstruation ceremony"
Impossible: "I saw three portions of blood."

(36) rê- s- anpun inta-ra.
1.SG.ERG 3.SG.ABS- saw rain–DUAL
"I saw two rain-men/men who came with the rain."

In the context of the generalization that for many speakers pluralization only applies to animate nouns, this strongly suggests that the plural operator presupposes that the noun it applies to is [+animate] and coerces an animate interpretation of the nouns in (35) and (36).[16] Crucially, as the examples in (33) and (34) show, numerals can modify mass nouns, in which case the mass noun denotes a plurality of countable portions, but pluralization itself does not induce a discrete portion reading.

As in Yudja, plural markers are restricted to animate nouns and, for some speakers, nouns denoting salient individuals. Note that in (32), the numeral may contribute to making individuals in the denotation of N salient and thus licensing the plural marker. It seems that portions per se are not salient enough individuals to meet the presuppositions of the plural operator.

6.3. Languages in Which Plurality Entails Discreteness (Portion Readings) but not Countability

We have seen so far that plurality does not presuppose discreteness. In English and Hebrew, we saw that both count and mass nouns can be pluralized, but that only (plural) count nouns necessarily denoted sums of sets of discrete entities. Count nouns are thus a subset of pluralizable nouns. In Yudja and Panará, in contrast, plurality presupposes animacy (or possibly a high degree of individuation), and a subset of countable nouns are pluralizable. In this section, we

[16] Fred Landman (p.c.) drew my attention to some similarities between the coercion illustrated in (35) and (36) and the coercion to "associated animates" evidenced in plurals such as "The two ham-sandwiches haven't got their orders yet."

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examine a third group of languages where not all pluralizable nouns are countable, but where pluralization does entail discreteness.

Sakurabiat (Mekens) is a Tuparian language spoken by fewer than 100 people in the Rio Mequens Indigenous territory. It is endangered, with only fourteen fluent speakers. The data in this section come from Galucio and Costa (2020).

Numerals in Sakurabiat modify nouns and do not require plural marking.

(37) a. kie sakurap
one spider monkey
b. turu sakurap
two spider monkeys

With relatively solid substances, counting portions is possible:

(38) a. kie tometome
a puddle of mud
b. turu tometome
two puddles of mud

(39) a. kie yẽra
one meat
"one piece of meat"
b. turu yẽra
two meat
"two pieces of meat"

With granulated and powdered substances, an explicit container is required:

(40) a. *kie pereyã
one manioc flour
"one pot of manioc flour"
b. kie pereyã aap
one manioc flour pot

(41) a. *kie kuut
one salt
"one measure/recipient of salt"
b. kie kuut tek
one salt measure/container

With liquids, the norm is to use an explicit container phrase. If the numeral occurs directly with a noun denoting a liquid, then a contextually determined container is understood, implying a null container phrase. Free portion readings are not available.

(42) tuero "chicha;fermented.beverage"
a. *kie tuero
one chicha
"(there is) one pan of chicha"
b. kie kwãe tuero (piro)
one pan chicha (have)

(43) tuero obaat ese-i-a-t o-kupi
chicha three/many SOC-come-THV-PST FOC 1SG-sister
"My sister brought three pans of chicha"

With "water" the container is obligatory, and a numeral modifying the noun is interpreted as denoting "river".

(44) uku 'water; river'
a. kie kwãe uku piro
one pan water have
"there is one pan of water"
b. turu uku so-a-r-õt
two water see-THV-PST-I
ONLY: "I saw two rivers"
(Not: I saw two pans of water)

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The only example of a liquid that occurs without any classifier in the corpus was "blood". Galucio and Costa note that it is also the only liquid marked with a possessive.

(45) turu s-au
two 3- blood
"two pools of someone's blood"

Sakurabiat has a plural marker *-yat* which has two uses: it can be used as a plural (46) or to contrast a plural with a singular meaning, as in (47). It is optional in the presence of a numeral.

(46) Aramira woman
aramira-yat women, a group of women

(47) pedro-yat Pedro's family

The plural attaches to all nouns (except those denoting animals). Crucially it can attach to liquids and (in context) granulated substances with a portion reading. However, these nouns cannot be directly counted.

(48) uku-yat so-at-õt
water-PL see-PST-I
"I saw a lot of puddles of water" (not necessarily a lot of water)

(49) a. kuut so-a-r-õt etu-pi sayẽ
salt see-THV-PST-I basket-inessive AUX.distributed
"I saw salt in the basket"
b. kuut-iat so-a-r-õt etu-pi sayẽ
salt-PL see-THV-PST-I basket-inessive AUX.distributed
"I saw salt (in packages) in the basket"

The portion reading is obligatory, and the plural morpheme in (48) and (49) cannot be interpreted as an abundance reading.

Similar facts are reported for two Karib languages, Ye'kwana and Taurepang. Ye'kwana is spoken in northeast Brazil and Venezuela by around 6000 speakers, while Taurepang is spoken in roughly similar areas by around 670 speakers (Coutinho 2020a, 2020b).

Ye'kwana allows bare nouns (50) and has a plural marker *-komo/-chomo*, which attaches to all nouns except for nouns denoting animals (51):

(50) yanwa faduudu n-ame-i (Ye'kwana)
man banana 3-eat-PRP
"the/a/some man/men ate a/the/some banana(s)."

(51) yanwa-komo faduudu-komo n-ame-i-cho
man-PL banana-PL 3-eat-PRP-PL
"The/Some men ate the/some bananas."

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Pluralization of substance denoting nouns results in a free portion reading of mass nouns.

(52) Marcelo n-enö-I **yadaachi-chomo**
Marcelo 3-drink-PRP caxiri-PL
"Marcelo drank portions of caxiri."

(53) **munu-komo** nonoojo nato
blood-PL floor 3.cop
"There are many portions of blood on the floor."

Numeral expressions combine with nouns denoting sets of individuals. When the noun occurs with a numeral, it does not pluralize (54):

(54) Marcelo **äddwawä** faduudu n-ame-i
Marcelo three banana 3-eat-PRP
"Marcelo ate three bananas."

However, despite the fact that bare substance nouns can pluralize and denote sets of portions, they cannot occur with numerals without the intervention of a classifier:

(55) a. Isabella tuna-komo neneanä
Isabella water-PL saw
"Isabella saw portions of water."
b. *Isabella ädwawa tuna neneanä
Isabella three water Saw
c. Isabella ädwawa chäwötö tuna neneanä
Isabella three bowl water saw
"Isabella saw three bowls of water."

Taurepang shows a similar pattern with respect to the relation between plurality and counting. Nouns can occur bare, and the plural *-ton* attaches to all nouns except for nouns denoting animals. These are pluralized using a special plural *-damök*.

(56) a. kurai-ton ere'ma-'pö-da
man-PL see-PAST-ERG
"I saw the men."
b. öynö-ton ere'ma-'pö-da
pan-PL see-PAST-ERG
"I saw the pans."

(57) a. kachiri-ton ere'ma-'pö-da
caxiri-PL see- PAST-ERG
"I saw (the) portions of caxiri."
b. aro-ton ere'ma-'pö-da
rice-PL see- PAST-ERG
"I saw the portions of rice."

Coutinho clarifies that the plural gives a multiple of portions reading and is not an abundance plural. Unlike in Ye'kwana, numerals combine with nouns which are marked plural. (58) and (59) give examples of numerals occurring with plural human-denoting and artefact-denoting nouns:

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(58) Seuröwöne kurai-ton etopök wo'nan se'na
Three man-PL went hunting
"Three men went hunting."

(59) Sakörörö canau-ton komemasa'da iwentamato peda
Four canoe-PL built sell POS
"I built four canoes to sell."

However, as in Ye'kwana, numerals cannot be combined with substance denoting nouns, either in the bare or pluralized form (60–61). These constructions require container classifiers:

(60) *U'wi ainapö'ya seröwarö sakörörö
Manioc_flour toast today four

(61) *Sakörörö u'wi-ton ainapö'ya seröwarö
Four manioc_flour.PL toast today

(62) U'wi ainapö'ya seröwarö sakörörö enpakapö'ya
manioc_flour toast today four bowl
"Today I toasted four bowls of manioc flour."

What these data clearly show, then, is that plurality in these two Karib languages and in Sakurabiat induce portion readings, which from the context in which they were used are clearly discrete, but the data shows that this is not enough to allow counting. Thus, the pluralization operation applied to mass nouns results in a set of contextually salient discrete portions closed under sum, but this set is not countable. Plausibly, then, pluralization in these languages presupposes a contextually salient set of discrete portions, but this operation is distinct from the mass-to-count shift, which is either lexical or, as I assume, expressed via an implicit classifier, and which results in a set of discrete *and* countable nouns.

We see then in these languages that pluralization is independent of countability and of the count–mass distinction.

7. Discussion and Conclusions[17]

We began this paper by making reference to widely held assumptions that the singular–plural distinction and the count–mass distinction are closely related. The availability of plural morphology was taken to be a "signature property" of count–mass languages (Chierchia 1998a), and though the existence of languages such as Dëne Sųłiné (see example 24) showed that the

[17] [Editorial comment] Susan did tell me that the main thing that had to be done was to rewrite the concluding section, get rid of redundancies, sharpen the conclusion, etc. I have done a bit of that, but not beyond what was really there in the text.

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count–mass distinction need not be associated with a singular–plural distinction in mass nouns, Chierchia (2010) maintained both that plural morphology is not expected in languages without a count–mass distinction and that in count–mass languages the singular–plural distinction will target count nouns (although abundance readings of pluralized mass nouns may be available).

Chierchia (1998a) gives the rationale behind the relation between count nouns and pluralization as follows. Mass nouns are lexical plurals, i.e. they denote sets closed under sum. Mass-to-count shift, when it occurs, picks out a set of discrete atomic entities relevant for counting, and pluralization closes this set under sum.

The underlying idea, expressed in terms of atoms, seems highly plausible. Counting requires access to a set of discrete minimal elements, atoms, as expressed in a standard interpretation of a cardinality predicate such as "three", given in (63), in which the cardinality of x is 3 iff the set of atomic parts of x has a cardinality of 3.

(63) *three:* $\lambda x.|\{y: y \sqsubseteq_{ATOM} x\}| = 3$

It is natural therefore (though not necessary) to have a morphological mechanism which distinguishes between the set of atomic instances of N, derived via the count-to-mass operation, which must be accessed in the case of counting, and the set of sums of these entities derived via pluralization. Since mass nouns are not countable and their denotations do not allow a subset of discrete atomic instances to be determined, this contrast between singular and plural is irrelevant for mass nouns. Plural mass nouns, if they occur, either have an abundance interpretation or will receive count noun interpretations by coercion.

Our explorations in this paper have suggested that the contrast between sets of sums and sets of discrete minimal entities is orthogonal to the count–mass distinction, both in English and Hebrew and also in more underrepresented languages.

We have examined three groups of languages, and found that while in all of them plausibly a pluralized noun denotes a set closed under sum, the relation between pluralization and count nouns is different in each group. Our results are summarized in Table 3.1.

We see that in English and Hebrew, count nouns are a subset of nouns which can pluralize, but only pluralized *count* noun denotations are generated by a set of discrete elements. Pluralized mass nouns denote sums, with possibly an abundance interpretation, but these sums are not necessarily sums of discrete entities. Discreteness, including event context-dependent discreteness, is entailed only by countability.

In Yudja and Panará, pluralization applies to a subset of count nouns: in Yudja this is the set of [+human] nouns, while in Panará for some speakers this

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Table 3.1. *Comparison of three groups of languages with respect to countability, plurality and portion readings*

	Count–mass?	Plural mass nouns?	Portion readings for substance nouns?
English, Hebrew	Count–mass distinction: only some nouns occur with numerals.	General plural marker. Pluralized mass nouns (i.e. plural but not countable) do not presuppose discreteness.	Countable portion interpretations available through a restricted mass-to-count shift and through classifiers.
Yudja, Panará	All nouns occur with numerals.	No general plural marker. Plural marker occurs with [+human] (Yujda) and [+ animate] (Panará) nouns.	Countable portion readings available for all mass nouns with numerals and *many/big*. Portions readings, overriding individual entity readings, may be available for notional count nouns.
Sakurabiat, Ye'kwana, Taurepang	Count–mass distinction: only some nouns occur with numerals.	Plural markers available. Plural mass nouns presupposing discreteness are possible, but these are not countable.	Portion readings available by pluralizing mass nouns, but these are not countable.

is the set of [+animate] nouns, and for other speakers the set contains nouns denoting stable and salient individuals. In these languages, it seems that countability requires highly context-dependent discreteness, with portions of substances being countable, while pluralization is used to mark pluralities of highly stable individuals, who maintain their discreteness across contexts.

The third group includes Sakurabiat, Ye'kwana and Taurepang. These are the languages in which count nouns are a subset of the pluralizable nouns, but where pluralizations of nouns must denote sets of discrete minimal elements closed under sum. Crucially, although these minimal elements are discrete, they are not countable, and the plural noun cannot be combined with a numeral. Thus, in these languages the plural operator does presuppose a set of discrete minimal elements (which can be individuals or portions), where coercion induces a portion reading on substance denoting nouns, but unlike in English these portion readings are not countable. So, while (at least at first sight) all the conditions that would make nouns countable in English seem to be there, this is not enough in these languages to make the interpretations count.

In Ye'kwana, where numerals modify non-pluralized nouns, it would be possible to argue that this is because numerals do not have the same presuppositions as pluralization, and thus do not coerce a discrete reading on substance nouns, but in Sakurabiat (a Tupian language) and Taurepang (a Karib language), numerals may occur with plural nouns. This is shown in (58) and (59) above for Taurepang, and in (64) for Sakurabiat (from Galucio and Costa 2020):

(64) turu te aramira=yat se-suruka taose puuk kwak-so-ab=ese
two FOC woman=COL;PL 3C-run.PL.THV peccary sound-see-NMLZ=LOC
"Two women ran when they heard the pig."

Again, despite the fact that numerals in general can modify pluralized nouns in these languages, there are pluralized substance nouns with discrete portion interpretations which are not countable. This suggests that context-dependent discreteness is not sufficient to license countability; instead what is required for countability is probably something more like stable discreteness across contexts.

These data suggest that pluralization, i.e. the closure of an N denotation under sum, and countability (which requires discreteness) are conceptually distinct. Pluralization may, but need not, be presuppositional. In English and Hebrew it does not presuppose discreteness. In Sakurabiat, Ye'kwana and Taurepang it does presuppose discreteness, and in Yudja and Panará it presupposes a human or animate denoting noun. Coercion effects support this claim. However, as Sakurabiat, Ye'kwana and Taurepang show, satisfying a discreteness presupposition need not be sufficient to make a noun countable.

This shows that the derivation of count nouns is independent from pluralization, and that countability is a grammatical property, not an effect that is derived presuppositionally. Furthermore, it suggests that the role of context in deriving count nouns may differ from language to language. All in all, the present paper supports the claim that pluralization and countability are conceptually distinct and that the precise way in which they are connected varies across languages.